THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

WEDNESDAY, NOVEMBER 11, 2026

THE MORNING PICTURE

Google opens Gemini 4 Carbon to all paying users, Stripe closes its OpenRouter deal, and builders argue over who checks the agents' work

Two weeks after a limited release to cyber defenders, Google made Gemini 4 Carbon available to paying developers, businesses and Ultra subscribers. Sundar Pichai showed off its most striking job: agents that generated more than six million lines of machine-checked proofs about the code behind the Signal protocol. Demis Hassabis had it rebuild his 1990s game Theme Park.

Uncle Bob Martin spent Tuesday arguing that agents are powerful but "ultimately irresponsible," so people must test and verify their work. Aaron Levie predicted that swarms of agents are the next jump in output, and Guillermo Rauch said Vercel's security agent found 57,897 vulnerabilities in under a week. Stripe closed its OpenRouter purchase, Nvidia opened its GR00T 2 robot model, and Elon Musk said SpaceX ordered solar-cell factory gear for Starmind satellites. In the weekly Leaderboards, Anthropic's Claude Fable 5.2 takes three boards, and newly published weights make Kimi K3.1 Pro the strongest open model on three more.

I · GEMINI 4 CARBON

Google opens Carbon to all paying users and shows it proving Signal's code

The model Google first gave to a small group of cyber defenders is now in its API, its business products and its top consumer tier. Google's leaders led with formal proofs, an old video game and the chips underneath.

6M+	**~1%**	**$2 / $10**
Lines of machine-checked Lean proofs Carbon generated about Signal's protocol code	Share of the codebase that needed human review, Pichai said	Introductory price per million input / output tokens, rising to $4 / $20 after Nov. 30

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

Pichai announced the wider release at 11:04am EST:

> *"Gemini 4 Carbon is rolling out today. It's built for sustained, long-horizon tasks, reasoning deeply through complex software engineering, cybersecurity, and scientific discovery. Coming globally via the Gemini API in @GoogleAIStudio and Gemini for business, and for Ultra subscribers in @antigravity, the @GeminiApp, and AI Mode in Search."*
>
> SUNDAR PICHAI · @SUNDARPICHAI · 11:04AM EST · POST

Google's announcement, by Koray Kavukcuoglu, Google's chief AI architect, says Carbon went to cyber defenders through Google's Fairwind program two weeks ago. It says Carbon ranks first on CursorBench 4.0, Perplexity's WANDR and Zapier's AutomationBench at "more than 2X lower cost per task," and second on Artificial Analysis's new MKA benchmark at more than six times lower cost.

Pichai's biggest claim came at 12:28pm EST:

> *"Our teams put Gemini 4 Carbon to work on a major security challenge: mathematically proving the code underlying the Signal protocol. Using AlphaProof Code, @GoogleDeepMind researchers guided agents to verify the protocol down to compiled device code, closing the gap where real bugs hide."*
>
> SUNDAR PICHAI · @SUNDARPICHAI · 12:28PM EST · POST

He listed the results: more than six million lines of proofs in Lean, a language whose checker confirms every step; "years of expert manual verification compressed to just weeks"; and about 1% of the codebase needing human review. The Signal protocol encrypts messages in the Signal app and many other messengers. DeepMind's write-up says the work took four weeks and is the first time the implementation of a widely used protocol like this has been formally verified by any method. Google is open-sourcing the proofs.



Google's main campus in Mountain View, California, seen from the air in 2013. Carbon's launch demo was a 3D game of a pelican riding a bike around it.
PHOTO: AUSTIN MCKINLEY · CC BY 3.0 VIA WIKIMEDIA COMMONS

Demis Hassabis @demishassabis · CEO, Google DeepMind

A year ago Hassabis rebuilt a test version of Theme Park, his 1990s game, with Gemini 3. At 12:41pm EST he wrote:

> *"Our latest frontier model Gemini 4 Carbon is rolling out today! Amazing progress from Gemini 3 → Gemini 4 in less than a year... Carbon reverse engineers my original Theme Park binaries, adds a new DeepMind themed ride, and ports it to the web in one shot - kind of mindblowing to see!"*
> DEMIS HASSABIS · @DEMISHASSABIS · 12:41PM EST · POST

Amin Vahdat, Google's chief technologist for AI and infrastructure, added that Carbon was "trained 100% on Google TPUs," its Tensor Processing Unit chips.

II · AGENTS AND THEIR CHECKERS

"Verify, verify, verify," says Uncle Bob, as Levie bets on swarms

The day's sharpest argument was about checking agents' work, not building agents. Martin said someone must always take responsibility, Levie said swarms will change who wins, and Rauch put numbers on an agent that reviews code.

57,897	14,665	6,029
Vulnerabilities Vercel Security Review found in under a week, Rauch said	Of those already patched by Vercel's agent	Of those rated critical or high severity

Uncle Bob Martin @unclebobmartin · Author, Clean Code

The software author Dave Farley wrote that "AI that writes code faster is only useful if you can tell, just as fast, whether that code is any good." Martin, whose reply was his most-liked post of the day, answered at 11:14am EST:

> *"True. This does not imply that you have to review every line. Lots of systems have had full code reviews and still should not have been deployed. Test, test, test. Measure crap. Do mutation tests. Check dependencies. Look at the module structure. Use the agents to interrogate the system. Verify, verify, verify. The productivity win is big; but the risk is high unless you are very diligent in using the best tools at your disposal; efficiently enough to keep that win."*
> UNCLE BOB MARTIN · @UNCLEBOBMARTIN · 11:14AM EST · POST

CRAP is a code-risk score that combines complexity and test coverage; his tools for it are on GitHub.

A self-described independent researcher then called the "coordinated timing" of Linus Torvalds, Martin and other top programmers embracing AI coding "entirely suspicious."

Martin called it "a foolish conspiracy theory" at 9:38am EST: "The models/agents became remarkably useful last January. The difference was so stark that those of us who were dabbling started to do a lot more than dabble." Pressed again, he replied that agents "need supervision," and that "someone always has to take final responsibility — and agents simply cannot do that. They are ultimately irresponsible."

Aaron Levie @levie · CEO, Box

Matt Huang, founder of Paradigm and Tempo, described "swarm riders," people who command "large swarms of agents to productive ends" with "productive token spend that is 1000x-10000x higher than median software engineers." Levie quoted it at 12:18am EST:

> *"Agent swarms have the potential to drive yet another leap in output from AI. The impact will be a clear separation between teams that adopt this pattern of work and those that don't."*
> AARON LEVIE · @LEVIE · 12:18AM EST · POST

Guillermo Rauch and Elon Musk @rauchg · CEO, Vercel · @elonmusk

Rauch posted the tally at 11:15am EST: 57,897 vulnerabilities found and 14,665 "already patched by Agent."

A separate argument about languages started with Musk. In a thread about rewriting AI training and inference software in C and C++, he wrote before dawn Tuesday, "Real men manage their own memory." Later he added: "So easy to trigger Rust programmers. Rust is a good language, but C is what you use for optimal performance on bare metal. With great power comes great responsibility." Rauch quoted Musk at 9:06pm EST:

> *"Rust is a superb language and ecosystem, especially given it was invented pre agentic coding. With agents, new languages will blossom, even purpose-built ones for specific usecases, and discretionary C or even LLVM IR, assembly… AI is fluid. No more one-size-fits-all."*
>
> GUILLERMO RAUCH · @RAUCHG · 9:06PM EST · POST

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

Sweeney called the Unreal Engine 5.9 preview "a new age of game development" at 11:00am EST. Epic's release notes call it the first Unreal Engine release built primarily with agentic workflows. Agents wrote its new web-browser support. The preview came together in about a week for Epic's AI MegaJam, Nov. 12–15.

He also weighed in on the Independent Games Festival (IGF), where a group of jurors said they will not vote for any game that used generative AI "in any capacity," after the developer House House disclosed that under 0.1% of its game Big Walk's code was AI-written. Replying to a developer who said no studio can survive "without using agentic coding," Sweeney wrote at 7:50am EST: "This movement, if successful, will render all western indie developers unviable in the worldwide game business. They will destroy the thing they claim to love."

III · PLATFORMS AND MONEY

Stripe completes its OpenRouter purchase as Hugging Face passes 20 million builders

The plumbing of the AI economy changed hands and grew. OpenAI opened ChatGPT to businesses, and Meta's Muse got its own inbox.

~125%	300 PB	1.2B
Growth in OpenRouter's weekly tokens since Stripe announced the deal in August	Models, datasets and agent data hosted on Hugging Face, 9 PB added last week	Weekly ChatGPT users that OpenAI says businesses can now reach

Patrick Collison @patrickc · CEO, Stripe

Stripe announced that "OpenRouter is now part of Stripe," closing a deal agreed in August. OpenRouter routes developers' requests to hundreds of models from many providers. Collison quoted the news at 5:23pm EST:

> *"OpenRouter is serving ~125% more tokens per week than it was when we first announced this acquisition in August. We have a lot of ideas for how to integrate Stripe's and OpenRouter's capabilities in the months ahead."*
>
> PATRICK COLLISON · @PATRICKC · 5:23PM EST · POST

Clément Delangue @ClementDelangue · CEO, Hugging Face

Delangue posted that Hugging Face passed "20M AI builders" and "300PB of models, datasets, agent traces, and RL environments," with 9 petabytes added by users in a single week. "October was our biggest month ever in terms of DAUs, MAUs and revenue," he wrote, meaning daily and monthly active users.

Greg Brockman @gdb · President, OpenAI

Brockman urged companies to "claim your business on ChatGPT and engage with customers in new ways." OpenAI's beta lets businesses claim a profile, manage their information and see how ChatGPT's 1.2 billion weekly users engage with them. Eligible US online stores and service firms can also build a Business Agent to answer customers' questions, sponsor it, and connect it to Shopify.

Alexandr Wang and David Singleton @alexandr_wang · @dps · Meta

Meta's Muse agent got its own email address for paid subscribers in the US and Canada. Wang, head of Meta Superintelligence Labs, wrote "forward it stuff or add it to a thread and watch it go." Singleton added that it is "available to everyone very soon. You choose who it can contact. Muse can work from its own inbox, with your approval." Uber chief executive Dara Khosrowshahi said Uber Courier deliveries now work inside Muse, and Expedia joined as a connector.

Nvidia opens GRooT 2, and SpaceX orders solar-cell lines for Starmind

Nvidia released a robot model that learns from video of people at work, a Stanford robot challenge quadrupled its success rate, and Musk put a "sentient sun" label on a German equipment order.

100K hrs
Training data Nvidia released with GR00T 2, about 11 years of experience

48%
Full-task success in the 2026 BEHAVIOR Challenge, up from 12% a year ago

100 GW
Solar production equipment SpaceX's Singulus partnership is meant to support

Jim Fan @DrJimFan · Director of AI, Nvidia

Fan announced the model at 11:07am EST:

> *"Introducing GR00T 2: our most advanced, open System One model for robotics. It runs entirely on your robot's onboard compute, fully untethered, and you own every weight of it."*
> JIM FAN · @DRJIMFAN · 11:07AM EST · POST

He listed long tasks "chained over 20+ steps and 5+ minutes," such as chemistry experiments, and called the release "an open-source buffet": model weights, an agent framework, a simulation engine, hardware designs and "100K hours" of training data, with the recipes for learning across robot bodies "coming December." "Everything that moves will be autonomous," he wrote.



A Unitree G1 humanoid at a trade show in November 2024. Nvidia says its GR00T 2 Turbo runs on a G1 with an onboard Jetson Thor computer, responding in 124 milliseconds.
PHOTO: SAYANESY · CC0 VIA WIKIMEDIA COMMONS (CROPPED)

Nvidia's project page says GR00T 2 learned from 402,730 hours of first-person video of people at work; the 100,000 released hours are a public slice of it. On 50 tasks the robot had never practiced, it succeeded 40% of the time with no extra training. Five demonstrations per task and a 30-minute fine-tune on one RTX 5090 graphics card lifted that to 95%.

Fei-Fei Li @drfeifei · Co-founder and CEO, World Labs

Li hailed the "incredible progress of robotic learning" in the second BEHAVIOR Challenge, run by her Stanford lab. Its results show 100 household tasks across seven scenes, up from 50 tasks in three scenes, with 265 participants against 96 a year ago, and full-task success up from 12% to 48%. Huawei's Noah EAI team won.

Elon Musk · Starmind solar @elonmusk · CEO, SpaceX and Tesla

SpaceX said it signed "a major supply contract with SINGULUS" for advanced solar-cell manufacturing "to support more than 100 GW of equipment as we accelerate the production of our Starmind satellites in Bastrop, Texas." Musk quoted it at 6:47pm EST:

> *"SpaceX has ordered 100 GW/year of HJT solar manufacturing equipment from Singulus in Germany. This is for Starmind to create a sentient sun."*
> ELON MUSK · @ELONMUSK · 6:47PM EST · POST

HJT means heterojunction, a cell design that layers thin amorphous silicon on a crystalline wafer. Singulus says it will deliver a turnkey heterojunction line to Bastrop starting in the first half of 2027, scaling to support more than 100 GW of production equipment. Its nine-month report puts the first line at more than €60 million. When one user joked about a "Dyson Swarm Speedrun," Musk replied "Pretty much."

Separately, Starlink introduced Mini 2, a portable dish with a quick-swap battery good for "up to 4.5 hours." Musk called it "Portable, lightning-fast Internet that works anywhere on Earth where Starlink is legal!"

QUICK HIT

Elon Musk on SoftBank's Masayoshi Son: "I remember that time when Masa, instead of putting $12B into Tesla for a 1/4 of the company to help me take Tesla private, instead invested the money in WeWork. That was dumb."

Claude Fable 5.2 takes three boards; Kimi K3.1 Pro narrows the open gap

Anthropic's new model leads WebDev Arena, the Artificial Analysis index and LiveBench. Moonshot AI's published weights make Kimi K3.1 Pro the best open model on three boards, but closed models still hold first place on all seven capability boards.

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Capability boards where a closed model holds first place

64.6
Claude Fable 5.2 (Max) on the Artificial Analysis index, against 46.5 for Kimi K3.1 Pro

18.8T
Tokens Tencent's new Hy4 processed on OpenRouter in its first week, second overall

The gap best closed vs. best open-weight

The open-weight gap is narrowest on LiveBench 2.0, where DeepSeek V4.1 Flash 1017 sits 11th, 14.6 points behind Claude Fable 5.2, and in Agent Arena, where Kimi K3.1 Pro trails Claude Opus 5.5 by 14.5 points of net improvement. On the Artificial Analysis index the gap fell to 18.1 points, from 27.4 a week ago, because Kimi K3.1 Pro now counts as open. It is widest in head-to-head web building, where Kimi K3.1 Pro is 148 rating points behind Fable 5.2, and on Humanity's Last Exam, where Kimi K2.5 trails GPT-6 Astra by 30.4 points.

Movers this week

Anthropic released Claude Fable 5.2 on Monday, and it is already first on WebDev Arena (1,836), the Artificial Analysis index (64.6) and LiveBench (84.9), and fourth on ARC-AGI-2 (93.8%). It has no Text or Agent Arena score yet. Gemini 4 Carbon, which led two boards last week, is now second on Artificial Analysis and has its first scores on ARC-AGI-2 (94.2%, third) and LiveBench (83.5, fourth). Moonshot AI's Kimi K3.1 Pro weights are on Hugging Face under the company's own license, and Artificial Analysis now labels the model open. Its OpenRouter traffic rose 46% on the week, to 4.42 trillion tokens. Tencent's Hy4 entered OpenRouter's chart at No. 2.

BOARD · AS OF	LEADER	SCORE	BEST OPEN-WEIGHT MODEL	RANK	SCORE	GAP
LMArena Text · Nov. 9	Gemini 4 Carbon (High) · Google	1,526	Kimi K3 (Max) · Moonshot AI	19	1,487	39
LMArena WebDev · Nov. 10	Claude Fable 5.2 (xHigh) · Anthropic	1,836	Kimi K3.1 Pro (Max) · Moonshot AI	11	1,688	148
LMArena Agent · Nov. 10	Claude Opus 5.5 (Max) · Anthropic	+19.5%	Kimi K3.1 Pro (Max) · Moonshot AI	14	+5.0%	14.5
Artificial Analysis v5 · read Nov. 11	Claude Fable 5.2 (Max) · Anthropic	64.6	Kimi K3.1 Pro · Moonshot AI	10†	46.5	18.1
ARC-AGI-2 · Nov. 10	GPT-6 Astra (Max) · OpenAI	95.0%	DeepSeek V4.1 Flash 1017 (Max) · DeepSeek	21†	78.1%	16.9
LiveBench 2.0 · Nov. 6	Claude Fable 5.2 · Anthropic	84.9	DeepSeek V4.1 Flash 1017 · DeepSeek	11	70.3	14.6
Humanity's Last Exam · read Nov. 11	GPT-6 Astra · OpenAI	54.8%	Kimi K2.5 · Moonshot AI	17	24.4%	30.4
OpenRouter usage · Nov. 10	DeepSeek V4.1 Flash 1017 · DeepSeek (open)	53.8 trillion	Most-used closed model: Hy4 from Tencent, second, 18.8 trillion tokens for the week			

Rank is the board's own position; † counts each model once at its best setting (ARC-AGI-2 excludes its human panel). Gap is in each board's units: rating points for the Text and WebDev arenas, index points for Artificial Analysis and LiveBench, percentage points for Agent Arena net improvement, ARC-AGI-2 and Humanity's Last Exam. Open-weight status follows each board's license label or the model's published weights. LMArena lists Hy4 as proprietary; only its preview has published weights. Artificial Analysis and Humanity's Last Exam show no update date.

#	LMARENA TEXT NOV. 9	LMARENA WEBDEV NOV. 10	LMARENA AGENT NOV. 10	ARTIFICIAL ANALYSIS V5 READ NOV. 11	ARC-AGI-2 NOV. 10	LIVEBENCH 2.0 NOV. 6
1	Gemini 4 Carbon (High) Google · 1,526	Claude Fable 5.2 (xHigh) Anthropic · 1,836	Claude Opus 5.5 (Max) Anthropic · +19.5%	Claude Fable 5.2 Anthropic · 64.6	GPT-6 Astra OpenAI · 95.0%	Claude Fable 5.2 Anthropic · 84.9
2	Gemini 4 Argon (High) Google · 1,522	Claude Opus 5.5 (Max) Anthropic · 1,814	Gemini 4 Carbon (High) Google · +16.8%	Gemini 4 Carbon Google · 60.0	GPT-6.1 Sol OpenAI · 94.2%	GPT-6 Astra OpenAI · 84.5
3	Muse Wonder (Max) Meta · 1,511	Claude Sonnet 5.5 (xHigh) Anthropic · 1,773	Claude Sonnet 5.5 (Max) Anthropic · +16.1%	Claude Opus 5.5 Anthropic · 59.1	Gemini 4 Carbon Google · 94.2%	Claude Fable 5.1 Anthropic · 84.0
4	Claude Opus 5.5 (High) Anthropic · 1,506	Gemini 4 Carbon (High) Google · 1,768	Claude Fable 5.1 (Max) Anthropic · +14.5%	GPT-6 Astra OpenAI · 56.9	Claude Fable 5.2 Anthropic · 93.8%	Gemini 4 Carbon Google · 83.5
5	Claude Fable 5 (High) Anthropic · 1,504	GPT-6 Astra (Max) OpenAI · 1,764	GPT-6 Astra (Max) OpenAI · +14.2%	GPT-6.1 Sol OpenAI · 54.6	Claude Opus 5.5 Anthropic · 93.3%	Claude Opus 5.5 Anthropic · 83.4

Each model counted once at its best setting. Every top-five place on every board is held by a closed model. Gemini 4 Carbon's arena scores are still marked preliminary.

Where the leading open-weight models stand

MODEL	TEXT	WEBDEV	AGENT	ARTIFICIAL ANALYSIS	ARC-AGI-2	LIVEBENCH 2.0	OPENROUTER WEEK
Kimi K3.1 Pro Moonshot AI	24 · 1,483	11 · 1,688	14 · +5.0%	46.5	—	13 · 68.7	9 · 4.42
DeepSeek V4.1 Flash 1017 DeepSeek	—	17 · 1,664	22 · +2.1%	32.6	78.1%	11 · 70.3	1 · 53.8
Kimi K3 Moonshot AI	19 · 1,487	18 · 1,658	26 · +0.6%	—	60.4%	18 · 65.0	—
MiMo-V2.6-Pro Xiaomi	30 · 1,480	27 · 1,624	18 · +2.3%	29.9	—	17 · 66.1	—
GLM-5.3 Flash Zhipu	40 · 1,474	35 · 1,609	37	27.8	65.8%	22 · 60.7	4 · 11.3

Cells show rank · score (OpenRouter: rank · tokens in trillions, week to Nov. 10; Agent: rank · net improvement). Artificial Analysis shows index v5 scores; ARC-AGI-2 shows the semi-private score. A dash means the model is not listed or not in the board's top 10 for usage.

What each board measures: LMArena Text and WebDev rate models from user votes; Agent scores real agent sessions by net improvement. Artificial Analysis combines evaluations it runs itself. ARC-AGI-2 (Abstraction and Reasoning Corpus for Artificial General Intelligence) tests reasoning on novel puzzles. LiveBench 2.0 grades real work such as decks, reports and bug fixes. Humanity's Last Exam scores expert-written questions. OpenRouter ranks models by tokens processed, a measure of use, not quality.

Takeaways

1. Google's showcase for Carbon was proof, not a benchmark: millions of machine-checked lines about Signal's code.
2. The coding debate has moved to who checks the agents; Uncle Bob Martin's answer is tests, mutation testing and a responsible human.
3. Swarms are the next pitch, from Google's /swarm to Levie's forecast and Epic's agent-built Unreal Engine preview.
4. Stripe now owns OpenRouter, whose weekly tokens more than doubled since August.
5. Closed models lead every capability board, but Kimi K3.1 Pro cut the open gap on Artificial Analysis to 18.1 points.

On the beat

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Elon Musk, Uncle Bob Martin, Demis Hassabis, Patrick Collison, Palmer Luckey, Tobi Lütke, David Heinemeier Hansson, Fei-Fei Li, Greg Brockman, Mustafa Suleyman, Paul Graham, Clément Delangue, Jim Fan, Alexandr Wang, David Singleton, Lisa Su, Sundar Pichai, Joe Gebbia, Kanjun Qiu, Guillermo Rauch, Michael Dell, Aaron Levie, Tim Sweeney, Suhail Doshi, Amin Vahdat.

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Marc Andreessen, Jensen Huang, Ashok Elluswamy, Andrej Karpathy, Sam Altman, Dario Amodei, John Carmack, Yann LeCun, Satya Nadella, Vitalik Buterin, Jeff Dean, François Chollet, Andrew Ng, Michael Truell, Tom Brown, Mira Murati, Mark Zuckerberg, Noam Brown, Ilya Sutskever, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen, Kevin Weil, Aidan Gomez, Albert Gu, Richard Sutton.

Added Koray Kavukcuoglu (@koraykv), senior vice president at Google DeepMind and Google's chief AI architect. The list now stands at 53.

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